



Network Swiss Army Knife (NSAK)

Bachelor Thesis – Integrating Agentic AI for Network Reconnaissance

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Overview

- ▶ Introduction
- ▶ Implementation
- ▶ Evaluation Setup
- ▶ AI Integration
- ▶ Results
- ▶ Discussion and Conclusion
- ▶ Demo: Agentic Reconnaissance in Action

Introduction

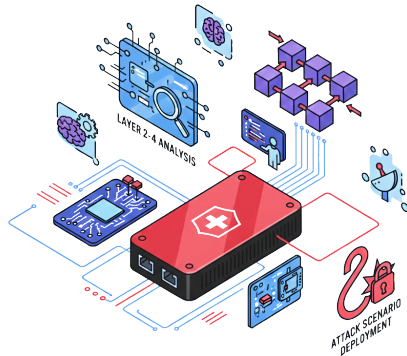
Starting Point: The NSAK Framework

Established in the predecessor project (Project 2)

A modular, open-source network security framework

Built around four reusable resource types:

- Device execution host at a network position
- Environment target network topology
- Scenario a red/blue team use case
- Drill reusable building block



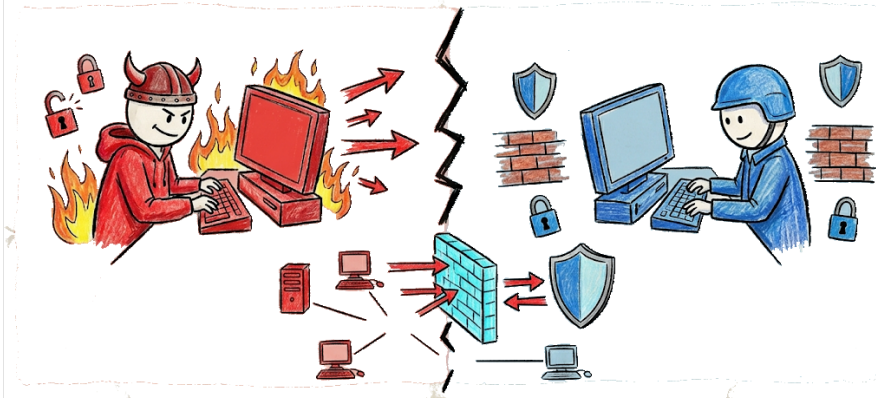
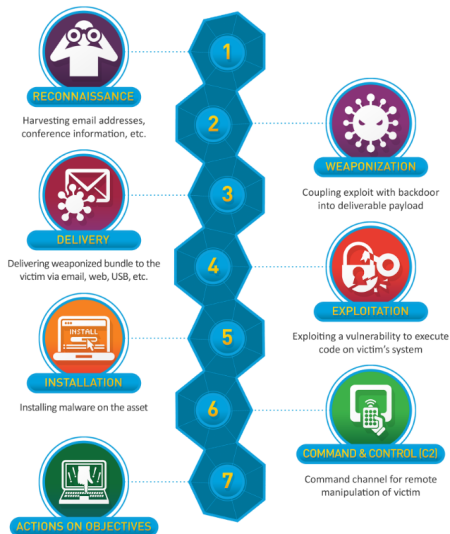


Figure: Red and Blue Team

Red Team



Blue Team

- Gather baseline data
- Detect
- Alert
- Investigate
- Plan a response
- Execute

Reconnaissance

The first stage of the cyber kill chain



Gathering information about a target network *before* any attack – it builds the map every later stage relies on, and is the focus of this thesis.

- 🔍 **Host discovery** *which systems are reachable on the network*
- 🔌 **Service & port enumeration** *what is running and exposed*
- 🏗️ **Characterization** *OS, versions and vulnerability indicators*

Hypothesis and Research Questions

From static scenarios to agentic AI

Hypothesis

Integrating agentic AI into NSAK improves adaptability, flexibility and automation over static scenarios, while lowering the operator's required expertise to interpret results and plan follow-up steps.

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- RQ1 How does an agentic AI scenario perform **compared to a static reference** scenario in network reconnaissance?
- RQ2 To what extent can agentic AI **support operators** during interactive reconnaissance?
- RQ3 To what extent is the implementation **suitable for real-world** deployment?

Project Goals

Order of implementation: Must → Should → Could

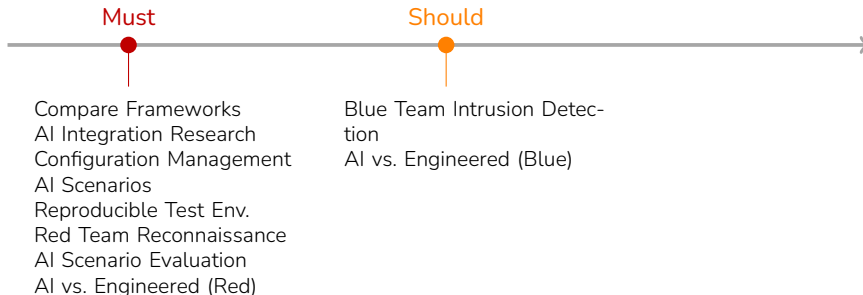
Must



- Compare Frameworks
- AI Integration Research
- Configuration Management
- AI Scenarios
- Reproducible Test Env.
- Red Team Reconnaissance
- AI Scenario Evaluation
- AI vs. Engineered (Red)

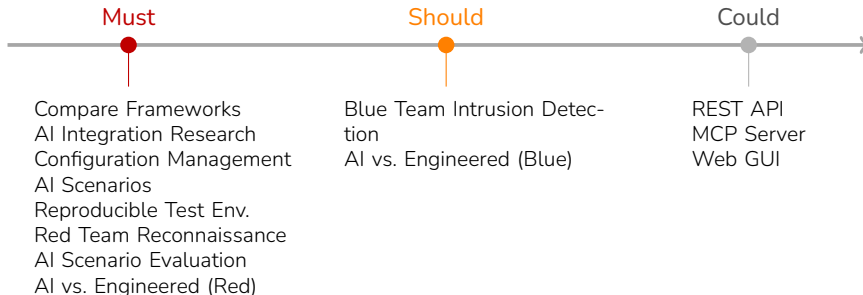
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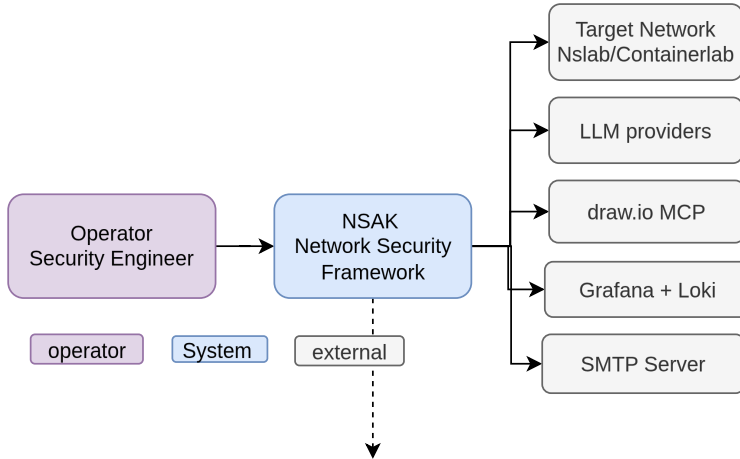
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Implementation

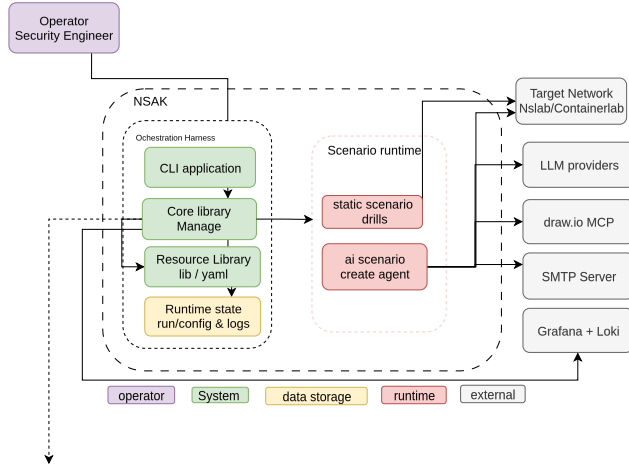
Layered Architecture

system-diagram



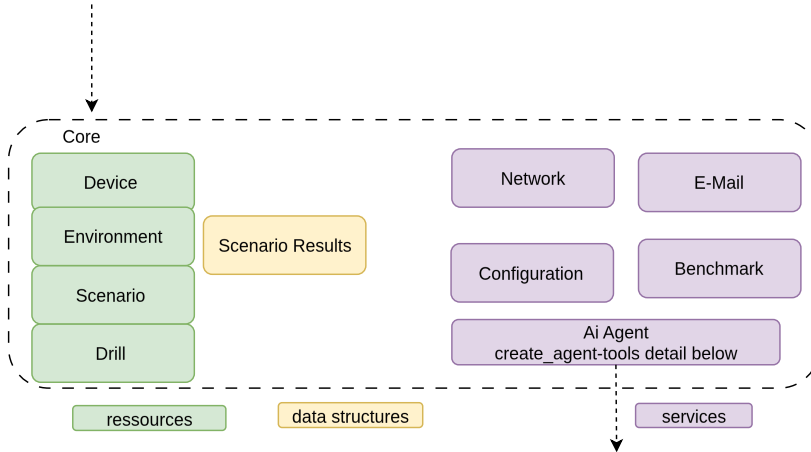
Layered Architecture

Lean Component Diagram



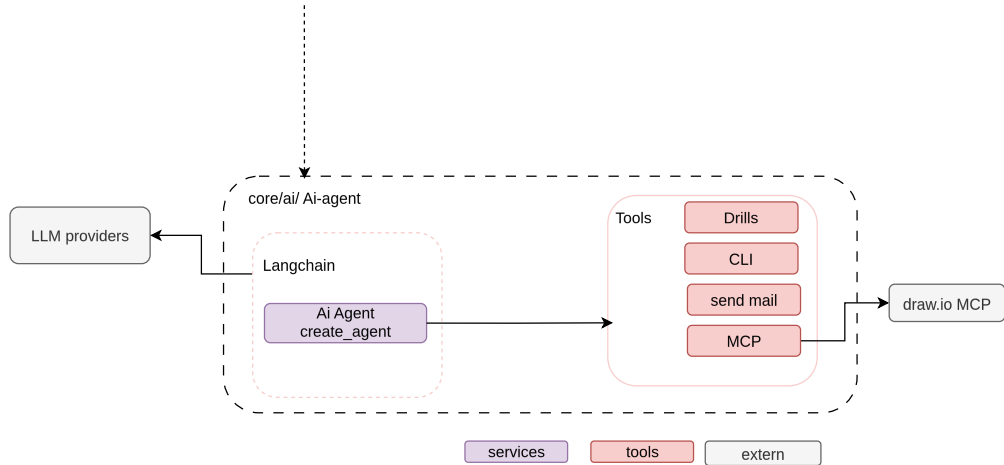
Layered Architecture

Core Diagram and Agent



Layered Architecture

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Evaluation Setup

Methology

3 LLMs • 2 Network Environments • Static Reference Baseline

LLMs

- qwen3: Self-hosted via Ollama
- gpt-oss: BFH-TI MLMP
- claude-opus-4-7: Anthropic API

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Network Environments

- Reproducible Containerlab Network
- BFH Network Security LAB

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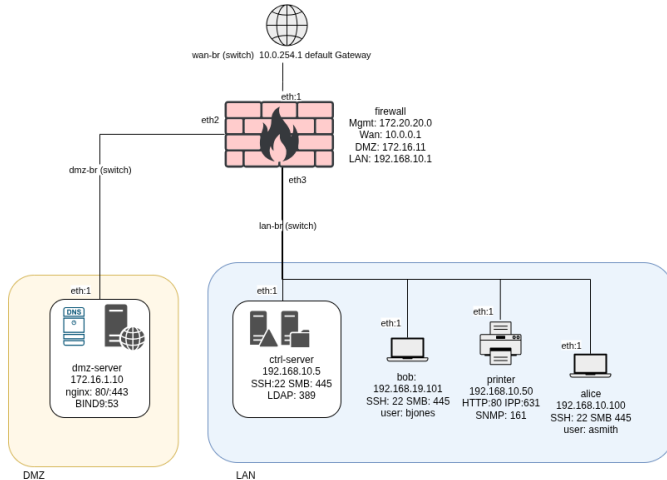
- Reproducible Containerlab Network
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Reconnaissance Scenario

- Accepts an interface and/or a subnet as arguments
- discover_network_interfaces: Iterates all interfaces if none is provided
- dhcp_request: Gets an IP via DHCP if needed
- discover_hosts: arp-scan & nmap
- enumerate_services: nmap NSE scripts
- Returns a ReconnaissanceScenarioResult

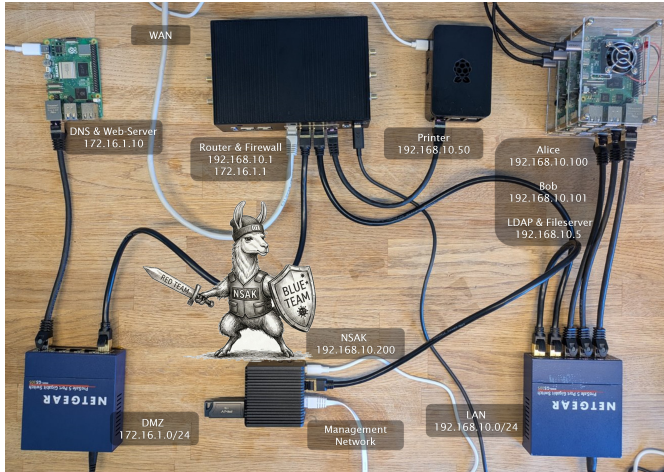
Reproducible Test Environment

A virtual enterprise network built with Containerlab



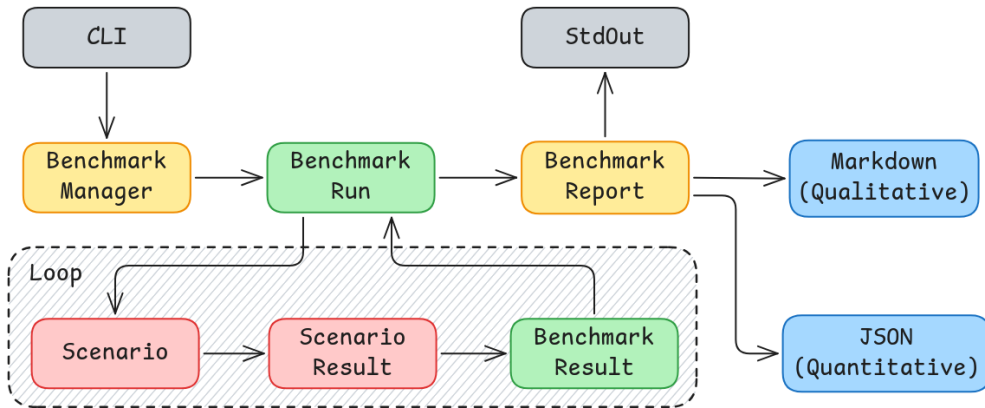
Physical Lab

The same network implemented with real hardware



Benchmarking Suite

Automated test runs for non-interactive scenarios



AI Integration

1. Provider Model und Agent
2. Tool calling (Langchain Tools) (MCP) (human interaction hook) (cli) (draw io) (send-mail)
3. 1. Single Agent interactive unstructured output
4. 2. Single Agent Reconnaissance unstructured output
5. 3. POC Blueteam intrusion detection - not followed time and scope - unstructured output
6. 4. Single Agent reconnaissance structure output (context-window, grammar -> why structured output create larger grammar)
7. 5. Multi Agent reconnaissance structured output (issue with smaller models) () Containerlab evaluation smaller model
8. 6. Multi Agent reconnaissance unstructured output

Concept 1: Taming the Context

From one overloaded agent to a focused pipeline

The problem

- One agent does everything in a single long conversation
- Every scan output piles up → the context keeps growing
- Smaller models lose the thread and start failing
- Cost and runtime become hard to predict

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- Split the task into **discovery** → **enumeration** → **assessment**
- Each phase is a **fresh agent** with its own, small context
- We pass on only the **distilled result**, not the whole history

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 Trade-off: Improves results when the context window is almost full, but adds orchestration overhead.

Concept 2: Keeping the Operator in Control

The agent proposes – the human decides on risky actions

Merge with previous slide

The problem

- An autonomous agent can run **any** command or send reports on its own
- In a security context that is unacceptable without oversight

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- The agent **pauses** before sensitive actions and asks for a decision
- Shell commands → **approve** before they run
- ✉ Sending a report → **approve / reject**
- 💬 Open questions → operator **responds** in free text

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🔄 After the decision the agent resumes exactly where it stopped – no restart, no lost progress.

Merge with previous slide

Single Agent

- One prompt to achieve the goal
- Long chain of tool calls, **growing context**
- Smaller models fail as context fills up

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Multi Agent *divide-and-conquer*

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- Smaller, focused context per step
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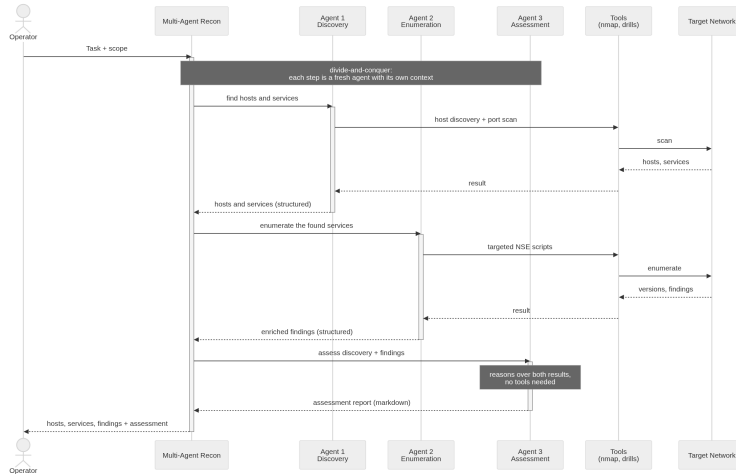
Structured vs. Unstructured output

Structured (schema-constrained): great for automated metrics, but **high failure rate** – especially for small models.

Unstructured: needed in the BFH lab where all models failed to produce valid structured output; freed the agent but required manual evaluation.

Multi-Agent Reconnaissance Pipeline

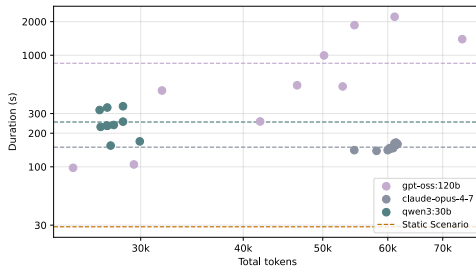
Discovery → enumeration → assessment, each a fresh agent



Results

Containerlab – Multi-Agent vs Single-Agent

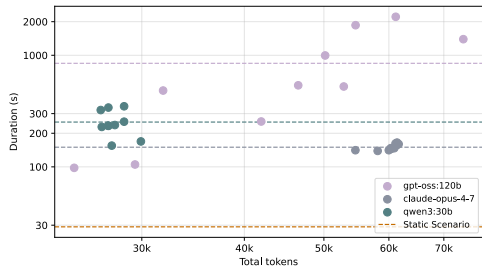
Tokens vs. duration – multi-agent first, then single-agent



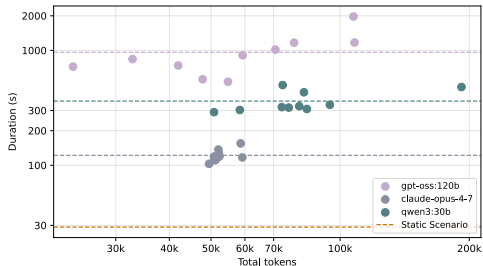
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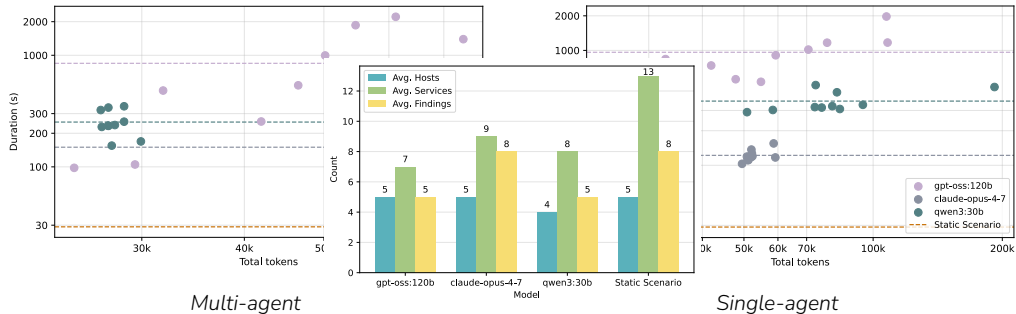
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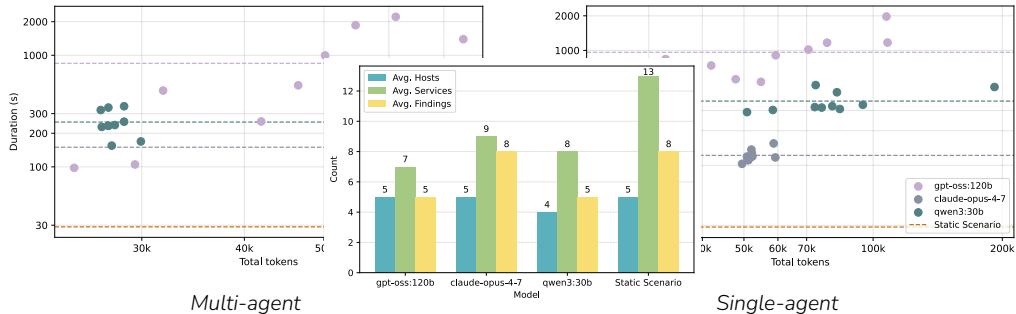
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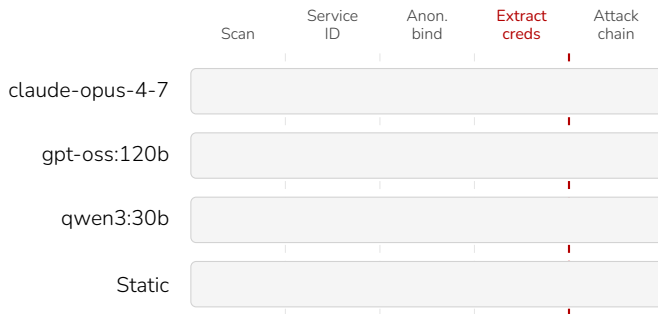
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- Decomposition: **qwen3:30b** drops tokens –68% and duration –30%.
- **claude-opus-4-7** keeps identical discovery quality (slight +13% token overhead).

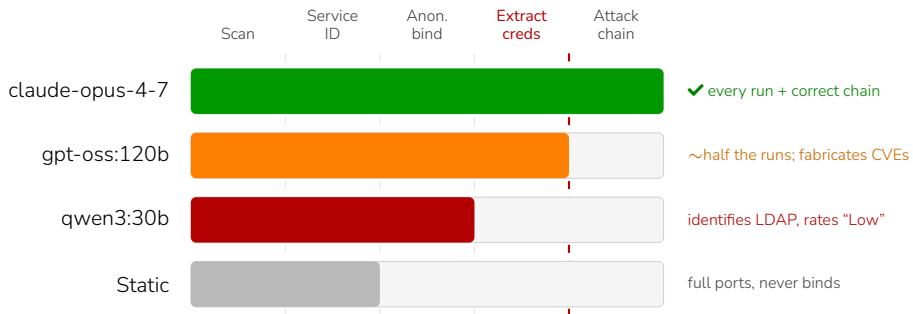
Qualitative Results + Interactive Mode

Q How far does each actor get along the attack chain?



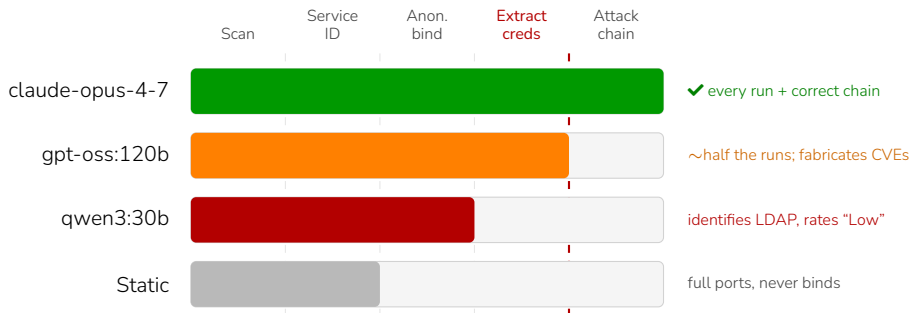
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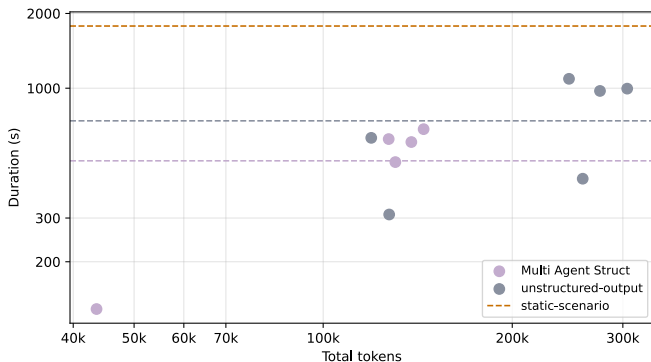
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Trade-off: static = breadth & determinism; AI = interpretive depth (severity-rated assessment, next steps).

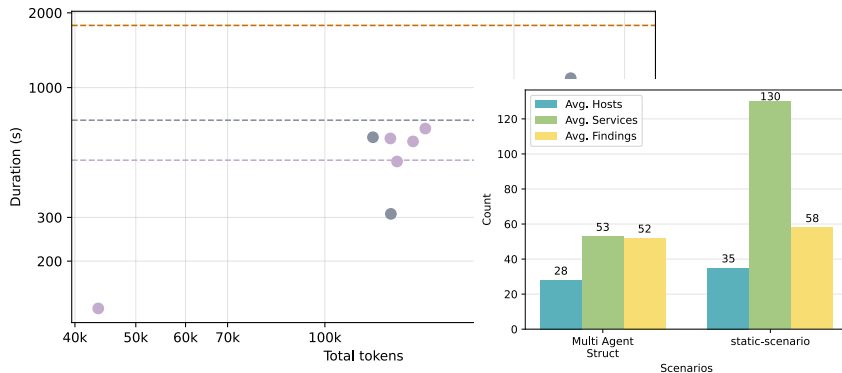
Network Security Lab – Multi-Agent Benchmark

NSLab - Findings



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Discussion and Conclusion

Key Findings

Takeaways across both environments

✓ What works

- Even smaller models can autonomously run **standard red-team reconnaissance**
- **Multi-agent decomposition** helps smaller models (-68% tokens for **qwen3:30b**)
- AI delivers **depth**: assessment, prioritized findings, remediation

Key Findings

Takeaways across both environments

✓ What works

- Even smaller models can autonomously run standard red-team reconnaissance
- Multi-agent decomposition helps smaller models (−68% tokens for `qwen3:30b`)
- AI delivers **depth**: assessment, prioritized findings, remediation

⚠ What limits it

- High failure rate – best-case subset
- Hallucinations: fabricated CVEs, honeypots

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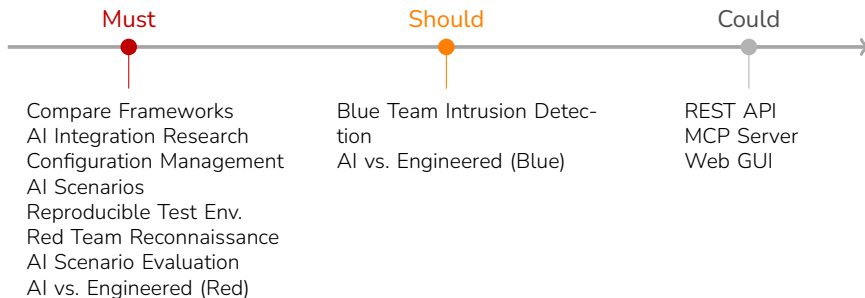
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- RQ3 **Deployment:** Scales to ~35 hosts, highlights key risks. **Requires supervision** (reliability limits).

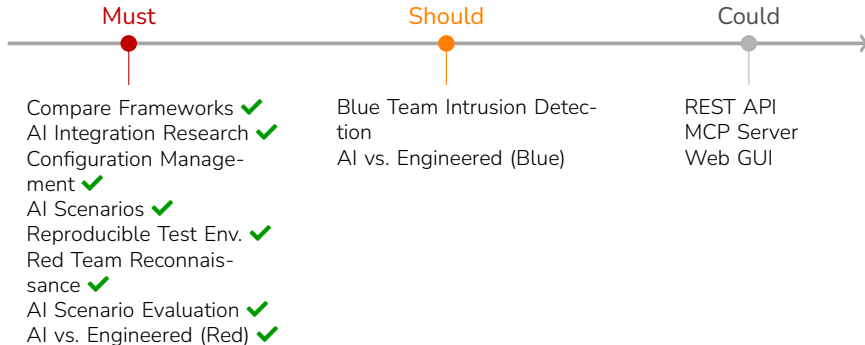
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Recommendations and Future Work

Recommendations

- Match model tier to task – frontier model for autonomous recon
- Multi-agent for smaller / self-hosted models
- Hybrid pipeline: static breadth + AI depth (RAG Retrieval-Augmented Generation)
- Human-in-the-loop + retry on incomplete runs

Future work

- Dynamic tool calling – expose only relevant tools
- Guardrails & sandboxing – scope restriction for production
- Robust HITL middleware
- Broader attack-chain coverage beyond reconnaissance

Conclusion

Agentic AI complements deterministic enumeration

- ✓ Using NSAK directly as an **agentic harness** proved effective – the agent autonomously and interactively executes network discovery via LangChain tools.
- ✓ Scripted scenarios excel in correctness and completeness; the agent adds **reasoning and assessment**, limited by model capability.
- ✓ **Multi-agent decomposition** models to digest large context improves models by reducing context complexity and increasing consistency.
- ⚠ Hallucinations remain a limitation – **supervised operation with a human reviewer** is the proposed deployment model.

LLMs make cybersecurity use cases more accessible and cost-effective – with a human in the loop.

Personal Conclusion

Frank

- ContainerLab impressed me – simple to deploy and reconfigure, motivated a home cyber lab
- Enjoyed researching state-of-the-art models, papers and LangChain
- Prefer LLMs mainly for reasoning and recommendations
- Surprised how hard autonomous NSAK deployment in the NSLab was
- Deep respect for academic writing – keeping a consistent “red thread” is hard

Lukas

- Initially skeptical of the usefulness of AI
- We covered a lot of topics we learned at BFH
- A bit of naivety in the planning led to a larger scope
- Keep focusing on the goals and research questions
- AI is here to stay as another engineering tool

Demo: Agentic Reconnaissance in Action